
Newsbot 2.0 – Technical Document

NATURAL LANGUAGE PROCESSING (ITAI-2373)
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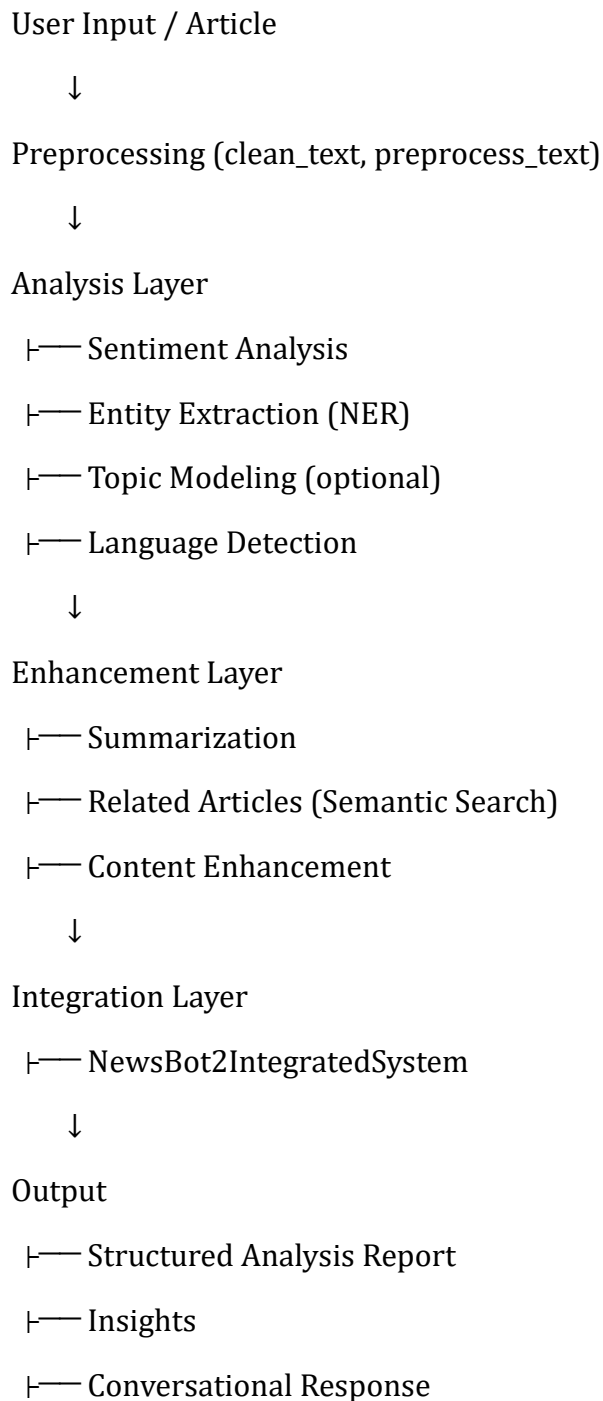
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1. Architecture Overview

a. System Design

NewsBot 2.0 follows a **modular, pipeline-based architecture** where each component performs a specific NLP task and passes results to the next stage.

Core Architecture Flow



b. Key Components and Interactions

Component	Role	Interaction
SentimentEvolutionTracker	Detects sentiment	Used in analysis + enhancer
EntityRelationshipMapper	Extracts entities	Used for insights + relationships
TopicDiscoveryEngine	Identifies themes	Optional (requires training)
IntelligentSummarizer	Generates summaries	Used in enhancer + reporting
SemanticSearchEngine	Finds similar content	Used in enhancer
MultilingualProcessor	Detects language	Used in analysis
ContentEnhancer	Combines outputs	Central intelligence layer
ConversationalInterface	Handles queries	User interaction layer
NewsBot2IntegratedSystem	Orchestrates everything	Main system controller

2. API Reference

a. Main System

i. NewsBot2IntegratedSystem

```
newsbot2 = NewsBot2IntegratedSystem()
```

ii. Methods:

- comprehensive_analysis(article_text)
 - Performs full NLP analysis
 - Returns: dictionary with sentiment, entities, summary, language, etc.

- `batch_analysis(articles)`
 - Processes multiple articles
 - Returns: list of analysis results
- `generate_insights_report(articles)`
 - Generates insights across articles
 - Returns: trends, summaries, insights

b. Content Enhancement

ContentEnhancer

- `enhance_article(article_text)`
 - Returns:
 - Entities
 - Sentiment
 - Topics
 - Related articles
 - Summary
- `generate_insights(articles)`
 - Returns:
 - Trends
 - Top entities
 - Combined summary

c. Summarization

IntelligentSummarizer

- `summarize_article(text, summary_type)`
 - Types: brief, balanced, detailed
- `summarize_multiple_articles(articles)`
 - Combines multiple articles into one summary

d. Sentiment Analysis

SentimentEvolutionTracker

- `analyze_sentiment(text)`
 - Returns:

- overall sentiment
- confidence score
- VADER breakdown

e. Entity Recognition

EntityRelationshipMapper

- `extract_entities(text)`
 - Returns named entities (people, orgs, locations)

f. Multilingual Processing

MultilingualProcessor

- `detect_language(text)`
 - Returns language + confidence
- `translate_text(text)`
 - Placeholder translation

g. Conversational Interface

ConversationalInterface

- `process_query(user_query)`
 - Detects intent + returns response
- `handle_follow_up(query)`
 - Maintains context

h. Testing Framework

NewsBot2TestSuite

- `run_all_tests()`
 - Runs:
 - component tests
 - integration tests
 - performance tests
 - edge case tests

i. Evaluation Framework

NewsBot2Evaluator

- `evaluate_classification_performance(data)`
- `evaluate_summarization_quality(data)`
- `evaluate_user_experience(queries)`
- `generate_evaluation_report()`

3. Installation Guide

Step 1: Install Dependencies

```
!pip install pandas numpy scikit-learn nltk spacy matplotlib seaborn langdetect
!python -m spacy download en_core_web_sm
```

Step 2: Download NLTK Data

```
import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('vader_lexicon')
```

Step 3: Load Dataset

```
df = pd.read_csv("BBC News Train.csv")
```

Step 4: Run Notebook Cells Sequentially

1. Setup & Imports
2. Preprocessing
3. Midterm Components (optional)
4. NewsBot 2.0 Components
5. Integrated System
6. Testing

Step 5: Run Test

```
result = newsbot2.comprehensive_analysis(sample_article)
```

4. Configuration Manual

1. Summarization Length

```
summarizer.summarize_article(text, summary_type="brief")
```


Options:

- brief
- balanced
- detailed

2. Topic Modeling (Optional)

`topic_engine = TopicDiscoveryEngine(n_topics=5)`

Change:

- number of topics
- training dataset

3. Sentiment Sensitivity

Uses VADER thresholds:

- Positive: > 0.05
- Neutral: -0.05 to 0.05
- Negative: < -0.05

4. Semantic Search Depth

`find_similar_articles(top_k=5)`

Change number of returned articles.

5. Language Detection

`multilingual.detect_language(text)`

Supports:

- English
- Spanish
- Arabic
- Urdu
- Hindi
- Chinese

6. Performance Optimization

- Limit article length (first 1000–1500 chars)

- Cache results in analysis_cache
- Use batch processing instead of single calls

7. Error Handling

System uses:

- try/except blocks
- fallback summaries
- default outputs

Summary

NewsBot 2.0 is a **modular, scalable NLP system** that:

- analyzes news articles
- extracts insights
- summarizes content
- supports multilingual processing
- enables conversational interaction

It is build on midterm foundations and extends them into a **complete intelligent news analysis platform**.